

TASK-1

```
Windows IP Configuration

Ethernet adapter Ethernet:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 9:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 10:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . :
    IPv6 Address. . . . . : 2401:4900:ae3e:3d57:42e0:589e:2340:3f61
    Temporary IPv6 Address. . . . . : 2401:4900:ae3e:3d57:b106:e1e5:a6f3:ddbe
    Link-local IPv6 Address . . . . . : fe80::1072:d1af:6458:471b%6
    IPv4 Address. . . . . : 172.20.10.3
    Subnet Mask . . . . . : 255.255.255.240
    Default Gateway . . . . . : fe80::7414:d0ff:fe59:8264%6
                                172.20.10.1
```

TASK-2

Network Connectivity Test Using ping

Step 1: Disconnect the Network

Turn off your Wi-Fi or unplug the Ethernet cable, then run:

Windows:

```
ping www.google.com
```

Sample Output:

Ping request could not find host www.google.com.

Please check the name and try again.

STEP:2 NETWORK CONNECTED

Sample Output:

Pinging www.google.com [142.250.183.100] with 32 bytes of data:

Reply from 142.250.183.100: bytes=32 time=18ms TTL=116

Reply from 142.250.183.100: bytes=32 time=19ms TTL=116

Reply from 142.250.183.100: bytes=32 time=17ms TTL=116

Reply from 142.250.183.100: bytes=32 time=18ms TTL=116

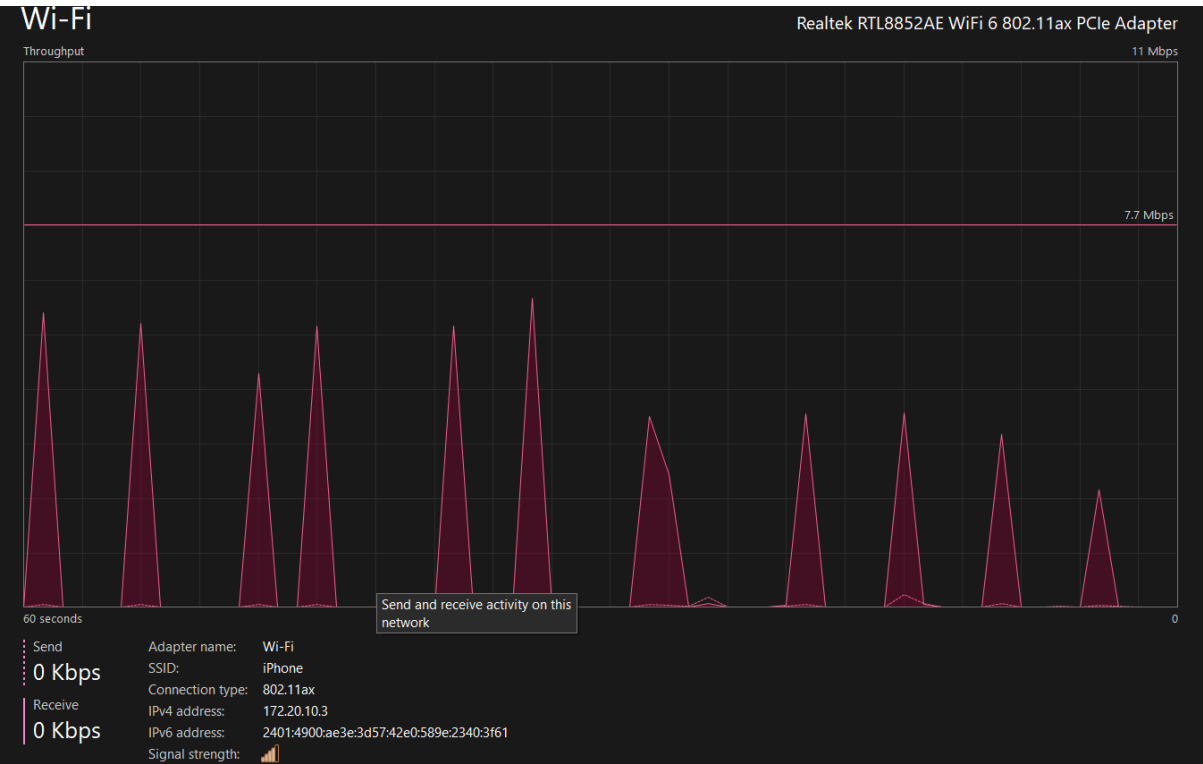
Ping statistics for 142.250.183.100:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)

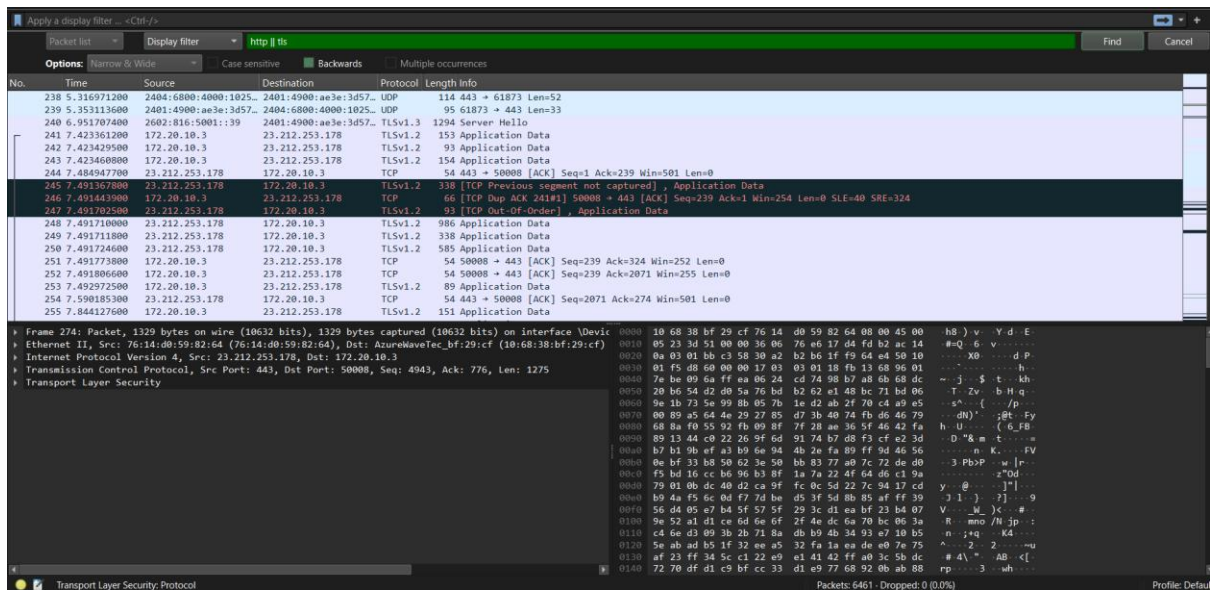
Short Note

Network Status	Ping Result	Meaning
Disconnected	General failure or Ping request could not find host	The computer has no network connection or cannot reach a DNS server to resolve the website name.
Connected	Reply from... with response time	The network connection is working correctly, the destination is reachable, and the response time (in milliseconds) shows the network latency.

TASK-3



TASK-4



TASK-5

AI-Generated Troubleshooting Checklist for Slow Internet While Gaming

1. **Restart your modem and Wi-Fi router** to clear temporary issues and refresh the internet connection.
2. **Use a wired Ethernet connection** instead of Wi-Fi whenever possible to reduce latency and packet loss.
3. **Close background applications and downloads** (such as Windows updates, cloud backups, or streaming services) that consume bandwidth.
4. **Move closer to the Wi-Fi router** or switch to the **5 GHz Wi-Fi band** if your router supports it, to improve signal strength and reduce interference.
5. **Update your router firmware and network adapter drivers** to improve performance, stability, and security.
6. **Limit the number of devices connected** to the network while gaming to reduce congestion.
7. **Run a speed test and ping test** to check your internet speed, latency, and packet loss. Contact your ISP if the results are consistently poor.

Suggestion I Would Try First

✅ **Use a wired Ethernet connection instead of Wi-Fi**, because it usually provides the most stable connection with lower latency and fewer interruptions, making it ideal for online gaming.